LESSON 02 · CHEAT SHEET

Decisions & Loops

IF / ELIF / ELSE

```
if months < 12:          # colon!
    rate = 0             # 4 spaces in
elif rating >= 4:        # only if above was False
    rate = 0.08
elif rating == 3:
    rate = 0.04
else:                    # none of the above
    rate = 0
print(rate)              # back out: after the if
```

First true branch runs, the rest are skipped. `elif` and `else` are optional.

CONDITIONS

`==` `!=`	equal, not equal (VBA: `=` `<>`)
`<` `<=` `>` `>=`	as in Excel
`a <= x < b`	between (chained)
`and` `or` `not`	combine
`text == "left"`	compare text, case matters

FOR: ONE ITEM AT A TIME

```
for name in "Orrin,Pell".split(","):
    print(name)          # Orrin, then Pell

for month in range(1, 13):
    print(month)         # 1 … 12  (13 not included)

for record in records.split(";"):
    name, sal, rat = record.split("/")
```

`range(5)`	0 1 2 3 4
`range(1, 6)`	1 2 3 4 5
`range(0, 20, 5)`	0 5 10 15

VBA `For Each x In …`. No counter, no `Next`.

WHILE: UNTIL SOMETHING CHANGES

```
years = 0
while salary < target:
    salary = salary * 1.03
    years += 1           # MUST move towards the end
print(years)
```

Runs for ever? **Ctrl+C** in the terminal. VBA: `Do While … Loop`.

THE LOOP RECIPE

```
total = 0                # 1. set up BEFORE
count = 0
highest = 0
for s in salaries:
    total += s           # 2. update INSIDE
    count += 1
    if s > highest:
        highest = s
print(total / count)     # 3. use AFTER
```

`x += 1`	`x = x + 1`
`x -= 1`, `x *= 2`	same idea

STEERING A LOOP

```
for x in items:
    if x == "left":
        continue         # skip to next item
    if x == "STOP":
        break            # leave the loop now
    ...
```

USEFUL IN CONDITIONS

```
day % 15 == 0            # every 15th: True on 15, 30
text.lower() == "yes"    # case-insensitive compare
if service == 1: unit = "year"
else:            unit = "years"
```

GOTCHAS

- Missing **colon** after `if` / `for` / `while` / `else`
- `if x = 3:` → `==`
- Mixed tabs/spaces: use 4 spaces, always
- `total = 0` inside the loop resets it every trip
- `range(1, 13)` stops at 12
- Checks in the wrong order: put the strictest first